Computer Science & Engineering

NIST University

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INSTITUTE PARK, PALLUR HILLS, BERHAMPUR, ODISHA -761 008

ASSIGNMENT NO # 01

Sub: Internet of Things — Due Date: *18*-02-2026

Program: B.Tech, 6th Semester (Batch: 2023) — Full Mark: 50

Q. No	CO	Level	**Answer all Questions. Each Question Carries 5 Mark** **BTL**: (1) Remembering (2) Understanding (3) Applying (4) Analyzing (5) Evaluating (6) Creating
1	1	2	Explain the Introduction to Internet of Things (IoT). Discuss the evolution of IoT and its major concepts.
2	1	2	Describe the Physical Design of IoT. Explain IoT devices, sensors, actuators, and communication protocols with examples.
3	1	2	Explain the Logical Design of IoT. Discuss IoT functional blocks and communication models.
4	1	2	Discuss various IoT Enabling Technologies such as RFID, Wireless Sensor Networks (WSN), Cloud Computing, and Big Data.
5	1	2	Explain different IoT Levels and Deployment Models. Compare cloud-based and edge-based deployment models.
6	1	2	Discuss any five domain-specific IoT applications such as Home Automation, Smart Cities, Agriculture, Healthcare, and Industry with suitable examples.
7	1	2	Explain the role of IoT in Smart Cities and Smart Agriculture. How does IoT improve efficiency and sustainability in these domains?
8	1	3	Discuss the major Challenges and Issues in IoT Systems, including security, privacy, scalability, interoperability, and data management.
9	2	2	Explain Machine-to-Machine (M2M) Communication. Differentiate between IoT and M2M with a comparison table.
10	2	2	Explain the concepts of Software Defined Networking (SDN) and Network Function Virtualization (NFV) in IoT. How do they improve network management and flexibility?

***CO1:**Students are able to understand the applications area of IoT, IoT enabling technologies, characteristics of IoT systems and IoT levels and Domain-specific IoTs.

***CO2:**Students are able to realize the revolutions of Internet in mobile devices, cloud and different sensor networks and apply them in IoT applications.